

PROJECT REPORT –

Project Title -

NexusHub – Connecting people through an intelligent event grid

1. Introduction

In modern community life, the ability to organize and discover events is essential. However, information is often fragmented across social media, emails, and physical posters. This lack of a central hub makes it difficult for organizers to reach their audience and for people to find events that match their interests.

NexusHub is a centralized platform designed to bridge this gap. It provides a professional ecosystem for organizers to host events of any kind—be it technical, social, cultural, or professional—and offers users a clean, "Bento-style" interface to discover and register for these experiences in one place.

2. Problem Statement

Most event discovery today happens on platforms not designed for it (like WhatsApp or Instagram), leading to:

- **Missed Opportunities:** Events are buried under unrelated content.
- **Low Engagement:** Organizers lack professional tools to manage RSVPs and track attendance.
- **Friction:** No easy way to save events to a calendar or get real-time updates on schedule changes.

NexusHub solves this by creating a dedicated, high-performance web application where every event has a home, and every user gets a personalized discovery experience.

3. Objectives of the Project

- **Centralization:** To provide a single platform for all local and regional events.
- **Efficiency:** To simplify the event creation process for organizers using AI-assisted tools.
- **Personalization:** To categorize events (Tech, Sports, Arts, etc.) so users only see what they care about.
- **Transparency:** To provide real-time RSVP counts and verified organizer profiles.

4. Requirement Gathering

Functional Requirements

1. **User Authentication:** Secure login for both Attendees and Organizers (Email/Google).
2. **Universal Event Creation:** Support for diverse event types with custom fields for date, time, venue, and description.
3. **Media Management:** Ability to upload event banners and PDF schedules/brochures.
4. **Discovery Dashboard:** A "Bento UI" that organizes events by category and popularity.
5. **RSVP & Notification System:** Users can register for events and receive automated reminders.
6. **AI Assistant (Side Bar):** An integrated agent to help organizers improve their event descriptions and select the right category.
7. **Search & Filtering:** Users can filter events by category, date range, or venue type.

Non-Functional Requirements

- **Responsiveness:** Must work seamlessly on mobile, tablet, and desktop.
- **Performance:** Minimal lag during image loading and dashboard transitions.
- **Scalability:** Capable of handling hundreds of concurrent RSVPs for popular events.
- **Reliability:** Data must be synced in real-time so users don't book filled events.

5. Software Requirements

- **Frontend:** React.js (for a fast, modern UI).
- **Styling:** Tailwind CSS (for a clean, dark-mode-first aesthetic).
- **Backend:** Firebase (Authentication and Firestore Database).
- **AI API:** Gemini 1.5 Flash (for the smart sidebar agent).

6. System Architecture

NexusHub uses a **Client-Server Architecture** optimized for real-time updates:

- **Presentation Layer:** Built with React, utilizing a Bento-box grid layout for high scannability.
- **Logic Layer:** Node.js functions and Gemini AI handle data validation and intelligent text generation.
- **Data Layer:** Firebase Firestore stores all event data, user profiles, and RSVP records.

7. Technology Stack

- **Frontend:** React.js
- **Styling:** Tailwind CSS
- **Database:** Firebase Firestore (NoSQL)
- **Auth:** Firebase Authentication
- **AI:** Gemini 1.5 Flash

8. Use Case Overview

User (Attendee):

- Browse events by category (Tech, Music, Food, etc.).
- RSVP and track upcoming events in a personal dashboard.
- Download event PDFs or schedules.

Organizer:

- Create and edit event listings.
- Use the AI sidebar to write better descriptions.
- View a list of registered attendees for each event.

Admin:

- Manage and verify organizers.
- Moderate event content to ensure community standards.

9. Future Scope

- **QR Code Check-in:** Generating unique codes for attendee verification at the venue.
- **Map Integration:** Showing event locations on an interactive local map.
- **In-App Messaging:** Allowing attendees to ask organizers questions directly.

10. Conclusion

NexusHub modernizes the way we organize and attend events. By moving away from scattered social media posts and into a dedicated, AI-enhanced hub, it ensures that local communities stay connected and that no great event goes unnoticed.

 ADMIN

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Resources

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Users

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Upgrade

SK

AllMembersGuestsInvited

Manage your team more effectively with new status and role columns, enhanced filtering, and improved sorting.

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Search by name or email

4 of 4 showingFilterSort

#	Name	Role	Status	Last activity	Invited by	Invited date
1	 Madhuri Farakate madhurifarakate399@gmail.com	Member	Active	Feb 13	Snehal Khot khot6834@gmail.com	Feb 13
2	 priyasavate9@gmail.com priyasavate9@gmail.com	Member	Active	–	Snehal Khot khot6834@gmail.com	Feb 13
3	 rautmaithil99@gmail.com rautmaithil99@gmail.com	Member	Active	Today	Snehal Khot khot6834@gmail.com	Feb 13
4	 Snehal Khot khot6834@gmail.com	Member	Active	Today	Joined independently	–

Compose

Inbox632

Starred

Snoozed

Sent

Drafts13

Purchases24

More

Labels

Notes

asana

Monday - tasks due soon: 1 - My workspace

Asana

Monday: tasks assigned to you

My workspace

View in Asana

Due soon: 1

Tasks due soon

Create a login page, College ID Verification FeaturesFeb 20

1 of 9

Mon, Feb 16, 7:31 PM

ReplyForward